



MVIDCodeReader SDK (C#)

Developer Guide

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Contents

Chapter 1 Overview	1
1.1 Introduction	1
1.2 Development Environment	1
1.3 Update History	1
Chapter 2 Typical Applications	2
2.1 Acquire Camera Image and Read Barcode	2
2.2 Upload Local Image and Read Barcode	4
Chapter 3 API Reference	5
3.1 Common APIs	5
3.1.1 MVID_CR_GetVersion_NET	5
3.1.2 MVID_CR_CreateHandle_NET	5
3.1.3 MVID_CR_DestroyHandle_NET	6
3.2 Camera Settings APIs	6
3.2.1 MVID_CR_CAM_EnumDevices_NET	6
3.2.2 MVID_CR_CAM_EnumDevicesByCfg_NET	6
3.2.3 MVID_CR_CAM_BindDevice_NET	7
3.2.4 MVID_CR_CAM_BindDeviceByIP_NET	7
3.2.5 MVID_CR_CAM_BindDeviceBySerialNumber_NET	8
3.2.6 MVID_CR_CAM_SetIntValue_NET	8
3.2.7 MVID_CR_CAM_GetIntValue_NET	9
3.2.8 MVID_CR_CAM_SetFloatValue_NET	9
3.2.9 MVID_CR_CAM_GetFloatValue_NET	10
3.2.10 MVID_CR_CAM_SetBoolValue_NET	10
3.2.11 MVID_CR_CAM_GetBoolValue_NET	11
3.2.12 MVID_CR_CAM_SetEnumValue_NET	11
3.2.13 MVID_CR_CAM_SetEnumValueByString_NET	12

3.2.14 MVID_CR_CAM_GetEnumValue_NET	12
3.2.15 MVID_CR_CAM_SetStringValue_NET	13
3.2.16 MVID_CR_CAM_GetStringValue_NET	13
3.2.17 MVID_CR_CAM_SetCommandValue_NET	14
3.3 Functional APIs	14
3.3.1 MVID_CR_Process_NET	14
3.3.2 MVID_CR_CAM_RegisterImageCallBack_NET	15
3.3.3 MVID_CR_CAM_RegisterImageBufferCallBack_NET	15
3.3.4 MVID_CR_CAM_RegisterAllEventCallBack_NET	16
3.3.5 MVID_CR_StartGrabbing_NET	16
3.3.6 MVID_CR_CAM_SetImageNodeNum_NET	17
3.3.7 MVID_CR_CAM_SetImageOutPutMode_NET	17
3.3.8 MVID_CR_CAM_GetOneFrameTimeout_NET	18
3.3.9 MVID_CR_CAM_GetImageBuffer_NET	18
3.3.10 MVID_CR_CAM_RegisterPreImageCallBack_NET	19
3.3.11 MVID_CR_RegisterExceptionCallBack_NET	19
3.3.12 MVID_CR_SaveImage_NET	20
3.3.13 MVID_CR_GetImageFileData_NET	21
3.3.14 MVID_CR_RuleLoad_NET	21
3.3.15 MVID_CR_ScriptLoad_NET	22
3.3.16 MVID_CR_StopGrabbing_NET	22
3.4 Algorithm Settings APIs	22
3.4.1 MVID_CR_Algorithm_SetIntValue_NET	22
3.4.2 MVID_CR_Algorithm_GetIntValue_NET	23
3.4.3 MVID_CR_Algorithm_SetFloatValu_NET	23
3.4.4 MVID_CR_Algorithm_GetFloatValue_NET	24
3.4.5 MVID_CR_Algorithm_SetStringValue_NET	24
3.4.6 MVID_CR_Algorithm_GetStringValue_NET	25

Chapter 4 Data Structure and Enumeration	26
4.1 Data Structure	26
4.1.1 MVID_CAM_ENUMVALUE	26
4.1.2 MVID_CAM_FLOATVALUE	26
4.1.3 MVID_CAM_INTVALUE_EX	26
4.1.4 MVID_CAM_OUTPUT_INFO	27
4.1.5 MVID_CAM_STRINGVALUE	27
4.1.6 MVID_CAMERA_INFO	27
4.1.7 MVID_CAMERA_INFO_LIST	28
4.1.8 MVID_CODE_INFO	29
4.1.9 MVID_CODE_INFO_LIST	29
4.1.10 MVID_EVENT_OUT_INFO	29
4.1.11 MVID_IMAGE_INFO	30
4.1.12 MVID_IMAGE_OUTPUT_MODE	30
4.1.13 MVID_POINT_I	31
4.1.14 MVID_PROC_PARAM	31
4.2 Enumeration	32
4.2.1 MVID_CODE_TYPE	32
4.2.2 MVID_IMAGE_TYPE	32
4.2.3 MVID_IMAGE_OUTPUT_MODE	34
Appendix A. Appendixes	35
A.1 Algorithm Parameters	35
A.2 Error Code	39

Chapter 1 Overview

1.1 Introduction

The MVIDCodeReader SDK is a software development kit using C# language, which can cooperate with industrial cameras to acquire images with barcode information, recognize 1D barcodes or 2D barcodes, set parameters, filter barcodes, and so on.

1.2 Development Environment

The development environment of MVIDCodeReader SDK is shown in the table below.

Item	Required
CPU	Intel Pentium IV 3.0 GHZ and above
RAM	1 GB and above
Operating System	Microsoft® Windows XP (32-bit)/Windows 7 (32/64-bit)/Windows 10 (32/64-bit)
MVS	V3.0.0 and above



Note

- For the operating system of Microsoft Windows XP, you should install the Microsoft .NET Framework 3.5 Service Pack 1.
 - When installing dongle, you should insert iMVS1001, or iMVS6000 (CodeReader).
 - Support soft encryption, and certain types of cameras can be unencrypted.
-

1.3 Update History

Summary of Changes in Version 1.2.0_Apr. 2023

Added an API for reading and converting the image: ***MVID_CR_GetImageFileData_NET*** .

Summary of Changes in Version 1.1.0_Oct. 2019

New document.

Chapter 2 Typical Applications

2.1 Acquire Camera Image and Read Barcode

Steps

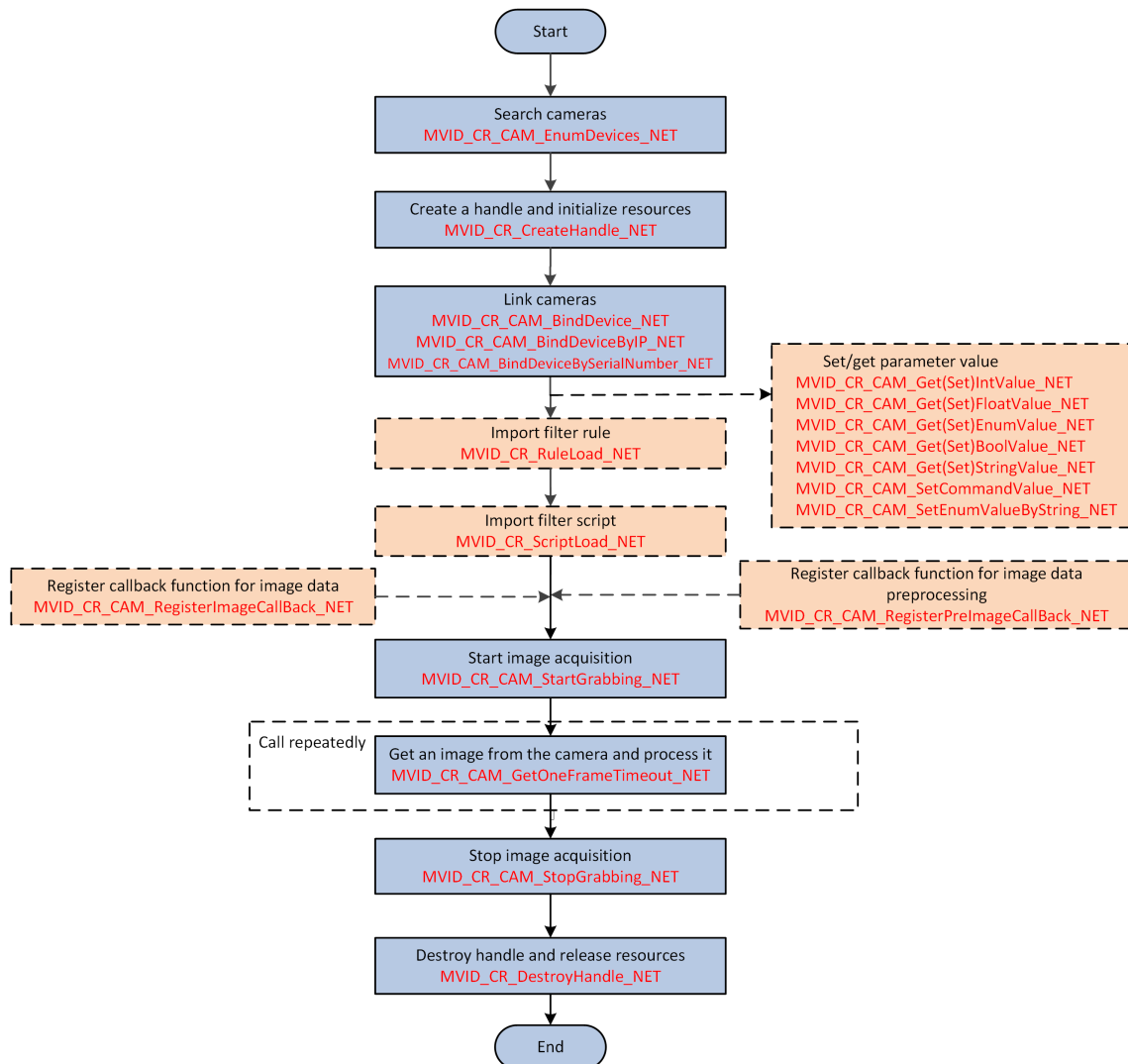


Figure 2-1 Programming Flow of Acquiring Camera Image and Reading Barcode

1. Call **MVID_CR_CAM_EnumDevices_NET** to search for devices.
2. Call **MVID_CR_CreateHandle_NET** to create the handle.
3. Link cameras.
 - Call **MVID_CR_CAM_BindDevice_NET** to link cameras via camera information.
 - Call **MVID_CR_CAM_BindDeviceByIP_NET** to link cameras via IP address.
 - Call **MVID_CR_CAM_BindDeviceBySerialNumber_NET** to link cameras via serial No.

4. Optional: Perform the following operation(s) to get or set different types of parameter values.

Parameters of type int	Call <u>MVID_CR_CAM_GetIntValue_NET</u> / <u>MVID_CR_CAM_SetIntValue_NET</u>
Parameters of type float	Call <u>MVID_CR_CAM_GetFloatValue_NET</u> / <u>MVID_CR_CAM_SetFloatValue_NET</u>
Parameters of type enum	Call <u>MVID_CR_CAM_GetEnumValue_NET</u> / <u>MVID_CR_CAM_SetEnumValue_NET</u> / <u>MVID_CR_CAM_SetEnumValueByString_NET</u>
Parameters of type boolean	Call <u>MVID_CR_CAM_GetBoolValue_NET</u> / <u>MVID_CR_CAM_SetBoolValue_NET</u>
Parameters of type string	Call <u>MVID_CR_CAM_GetStringValue_NET</u> / <u>MVID_CR_CAM_SetStringValue_NET</u>
Parameters of type command	Call <u>MVID_CR_CAM_SetCommandValue_NET</u>

5. Optional: Call [MVID_CR_RuleLoad_NET](#) to import filtering rule files.

6. Optional: Call [MVID_CR_ScriptLoad_NET](#) to import filtering script files.

7. Optional: Call [MVID_CR_CAM_RegisterPreImageCallback_NET](#) to register callback function for preprocessing image data.

8. Optional: Call [MVID_CR_CAM_RegisterImageCallback_NET](#) to register callback function for getting image data.

9. Call [MVID_CR_StartGrabbing_NET](#) to start image acquisition.

10. Call [MVID_CR_CAM_GetOneFrameTimeout_NET](#) repeatedly to get images from the camera.



Note

The API is invalid after calling [MVID_CR_CAM_RegisterImageCallback_NET](#). You should get the image data from the registered callback function (cbOutput).

11. Call [MVID_CR_StopGrabbing_NET](#) to stop image acquisition.

12. Call [MVID_CR_DestroyHandle_NET](#) to destroy the handle.

2.2 Upload Local Image and Read Barcode

Steps

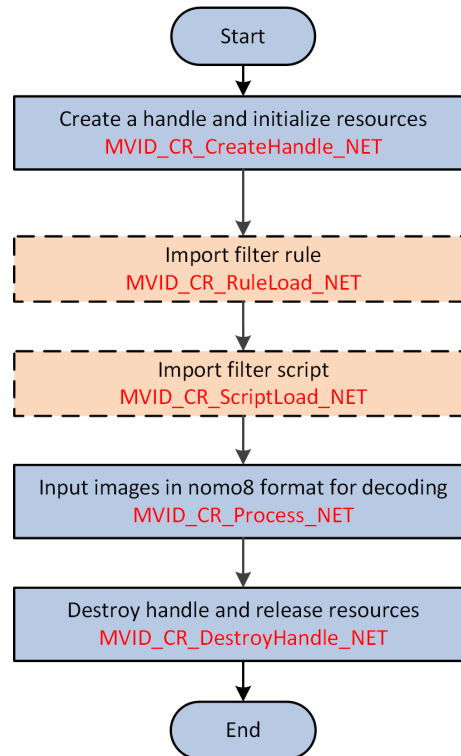


Figure 2-2 Programming Flow of Uploading Local Image and Reading Barcode

1. Call **MVID_CR_CreateHandle_NET** to create the handle.
2. **Optional:** Call **MVID_CR_RuleLoad_NET** to import filter rule files.
3. **Optional:** Call **MVID_CR_ScriptLoad_NET** to import filter script files.
4. Call **MVID_CR_Process_NET** to upload images in mono8 format for decoding.
5. Call **MVID_CR_DestroyHandle_NET** to destroy the handle.

Chapter 3 API Reference

3.1 Common APIs

3.1.1 MVID_CR_GetVersion_NET

Get version No. of SDK, 1D barcode, or 2D barcode.

API Definition

```
public static Int32 MVID_CR_GetVersion_NET(  
    IntPtr          pstrVersion  
);
```

Parameters

pstrVersion

Version No.

Return Value

Return version No. of SDK, 1D barcode, or 2D barcode for success. For example, 1.2.0.1_3.6.2_2.2.5.

3.1.2 MVID_CR_CreateHandle_NET

Create a handle.

API Definition

```
public Int32 MVID_CR_CreateHandle_NET(  
    UInt32          nCodeAbility  
);
```

Parameters

nCodeAbility

The capability set to be used. For example, you can call the command "MVID_BCR|MVID_TDCR" to enable functions of 1D barcode and 2D barcode.

Return Value

Return 0 for success, and return ***Error Code*** for failure.

Remarks

Do not connect the camera when creating a handle by this API.

3.1.3 MVID_CR_DestroyHandle_NET

Destroy handle and release resources.

API Definition

```
public Int32 MVID_CR_DestroyHandle_NET(
);
```

Remarks

Return *0* for success, and return ***Error Code*** for failure.

3.2 Camera Settings APIs

3.2.1 MVID_CR_CAM_EnumDevices_NET

Search for cameras on the current LAN.

API Definition

```
public static Int32 MVID_CR_CAM_EnumDevices_NET(
    ref MVID_CAMERA_INFO_LIST    pstCamList
);
```

Parameters

pstCamList

Device information list. See ***MVID_CAMERA_INFO_LIST*** for details.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.2 MVID_CR_CAM_EnumDevicesByCfg_NET

Search for specific cameras by configuration files.

API Definition

```
public Int32 MVID_CR_CAM_EnumDevicesByCfg_NET(  
    ref MVID_CAMERA_INFO_LIST    pstCamList  
);
```

Parameters

pstCamList

Camera information list. See [MVID_CAMERA_INFO_LIST](#) for details.

Return Value

Return 0 for success, and return [Error Code](#) for failure.

3.2.3 MVID_CR_CAM_BindDevice_NET

Link cameras.

API Definition

```
public Int32 MVID_CR_CAM_BindDevice_NET(  
    IntPtr    pstCamInfo  
);
```

Parameters

pstCamInfo

Device information.

Return Value

Return 0 for success, and return [Error Code](#) for failure.

3.2.4 MVID_CR_CAM_BindDeviceByIP_NET

Link a camera by IP address.

API Definition

```
public Int32 MVID_CR_CAM_BindDeviceByIP_NET(  
    String    chCurrentIp,  
    String    chNetExport  
);
```

Parameters

chCurrentIp

Camera's IP address

chNetExport

The IP address of the current PC.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.5 MVID_CR_CAM_BindDeviceBySerialNumber_NET

Link a camera by serial No.

API Definition

```
public Int32 MVID_CR_CAM_BindDeviceBySerialNumber_NET(  
    String      chSerialNumber  
);
```

Parameters

chSerialNumber

Camera serial No.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

Remarks

Only one camera can be bound via one handle.

3.2.6 MVID_CR_CAM_SetIntValue_NET

Set parameters of type int.

API Definition

```
public Int32 MVID_CR_CAM_SetIntValue_NET(  
    String      strKey,  
    Int64       nValue  
);
```

Parameters

strKey

Parameter name

nValue

Parameter value

Return Value

Return 0 for success, and return **Error Code** for failure.

3.2.7 MVID_CR_CAM_GetIntValue_NET

Get parameters of type int.

API Definition

```
public Int32 MVID_CR_CAM_GetIntValue_NET(  
    String          strKey,  
    ref MVID_CAM_INTVALUE_EX pIntValue  
);
```

Parameters

strKey

Parameter name.

pIntValue

Parameter value list, which contains the maximum values, minimum values, and current values of the parameters. See **MVID_CAM_INTVALUE_EX** for details.

Return Value

Return 0 for success, and return **Error Code** for failure.

3.2.8 MVID_CR_CAM_SetFloatValue_NET

Set parameters of type float.

API Definition

```
public Int32 MVID_CR_CAM_SetFloatValue_NET(  
    String          strKey,  
    Single          fValue  
);
```

Parameters

strKey

Parameter name.

fValue

Parameter value

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.9 MVID_CR_CAM_GetFloatValue_NET

Get parameters of type float.

API Definition

```
public Int32 MVID_CR_CAM_GetFloatValue_NET(  
    String          strKey,  
    ref MVID_CAM_FLOATVALUE    pFloatValue  
);
```

Parameters

strKey

Parameter name.

pFloatValue

Parameter value list. See ***MVID_CAM_FLOATVALUE*** for details.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.10 MVID_CR_CAM_SetBoolValue_NET

Set parameters of type boolean.

API Definition

```
public Int32 MVID_CR_CAM_SetBoolValue_NET(  
    String          strKey,  
    Boolean         bValue  
);
```

Parameters

strKey

Parameter name

bValue

Parameter value

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.11 MVID_CR_CAM_GetBoolValue_NET

Get parameters of type boolean.

API Definition

```
public Int32 MVID_CR_CAM_GetBoolValue_NET(  
    String          strKey,  
    ref Boolean     pBoolValue  
);
```

Parameters

strKey

Parameter name

pBoolValue

Parameter value

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.2.12 MVID_CR_CAM_SetEnumValue_NET

Set parameters of type enum.

API Definition

```
public Int32 MVID_CR_CAM_SetEnumValue_NET(  
    String          strKey,  
    UInt32         nValue  
);
```


Parameters

strKey

Parameter name

nValue

Parameter value

Return Value

Return 0 for success, and return **Error Code** for failure.

3.2.13 MVID_CR_CAM_SetEnumValueByString_NET

Set parameters of type enum by string.

API Definition

```
public Int32 MVID_CR_CAM_SetEnumValueByString_NET(  
    String    strKey,  
    String    sValue  
);
```

Parameters

strKey

Parameter name

sValue

Parameter value

Return Value

Return 0 for success, and return **Error Code** for failure.

Remarks

Machine vision cameras prior to V2.3.1 do not support the API.

3.2.14 MVID_CR_CAM_GetEnumValue_NET

Get parameters of type enum.

API Definition

```
public Int32 MVID_CR_CAM_GetEnumValue_NET(  
    String    strKey,
```

```
ref MVID_CAM_ENUMVALUE    pEnumValue
);
```

Parameters

strKey

Parameter name

pEnumValue

Parameter value list, which contains the current value and the number of valid values. See [MVID_CAM_ENUMVALUE](#) for details.

Return Value

Return *0* for success, and return [Error Code](#) for failure.

3.2.15 MVID_CR_CAM_SetStringValue_NET

Set parameters of type string.

API Definition

```
public Int32 MVID_CR_CAM_SetStringValue_NET(
    String          strKey,
    String          sValue
);
```

Parameters

strKey

Parameter name

sValue

Parameter value

Return Value

Return *0* for success, and return [Error Code](#) for failure.

3.2.16 MVID_CR_CAM_GetStringValue_NET

Get parameters of type string.

API Definition

```
public Int32 MVID_CR_CAM_GetStringValue_NET(
    String          strKey,
```

```
ref MVID_CAM_STRINGVALUE    pStringValue  
);
```

Parameters

strKey

Parameter name

pStringValue

Parameter value. See [MVID_CAM_STRINGVALUE](#) for details.

Return Value

Return *0* for success, and return [Error Code](#) for failure.

3.2.17 MVID_CR_CAM_SetCommandValue_NET

Set parameters of type command.

API Definition

```
public Int32 MVID_CR_CAM_SetCommandValue_NET(  
    String    strKey,  
);
```

Parameters

strKey

Parameter name

Return Value

Return *0* for success, and return [Error Code](#) for failure.

3.3 Functional APIs

3.3.1 MVID_CR_Process_NET

Read code.

API Definition

```
public Int32 MVID_CR_Process_NET(  
    IntPtr    pstParam,  
    UInt32    nCodeAbility  
);
```

Parameters

pstParam

Image information.

nCodeAbility

Code capability set

Return Value

Return 0 for success, and return **Error Code** for failure.

3.3.2 MVID_CR_CAM_RegisterImageCallBack_NET

Register callback function of image data.

API Definition

```
public Int32 MVID_CR_CAM_RegisterImageCallBack_NET(  
    cbOutputdelegate    cbOutput,  
    IntPtr              pUser  
);
```

Parameters

cbOutput

Image data callback function

pUser

User pointer

Return Value

Return 0 for success, and return **Error Code** for failure.

3.3.3 MVID_CR_CAM_RegisterImageBufferCallBack_NET

Register callback function for getting an original image.

API Definition

```
public Int32 MVID_CR_CAM_RegisterImageBufferCallBack_NET(  
    cbImageBufferdelegate    cbOutput,  
    IntPtr                  pUser  
);
```

Parameters

cbOutput

Original image function

pUser

User pointer

Return Value

Return 0 for success, and return **Error Code** for failure.

Remarks

The API is invalid after calling **MVID_CR_CAM_RegisterImageCallBack_NET**.

3.3.4 MVID_CR_CAM_RegisterAllEventCallBack_NET

Register callback function for getting camera events.

API Definition

```
public Int32 MVID_CR_CAM_RegisterAllEventCallBack_NET(  
    cbEventdelegate cbEvent,  
    IntPtr pUser  
);
```

Parameters

cbEvent

Callback function of camera events

pUser

User pointer

Return Value

Return 0 for success, and return **Error Code** for failure.

Remarks

The API should be supported by cameras.

3.3.5 MVID_CR_StartGrabbing_NET

Start image acquisition.

API Definition

```
public Int32 MVID_CR_StartGrabbing_NET(
);
```

Return Value

Return 0 for success, and return **Error Code** for failure.

3.3.6 MVID_CR_CAM_SetImageNodeNum_NET

Set the number of image cache nodes.

API Definition

```
public Int32 MVID_CR_CAM_SetImageNodeNum_NET(
    UInt32      nNum
);
```

Parameters

nNum

The number of cache nodes, which cannot be less than 1.

Return Value

Return 0 for success, and return **Error Code** for failure.

Remarks

The API should be called before acquiring images.

3.3.7 MVID_CR_CAM_SetImageOutPutMode_NET

Set image output format.

API Definition

```
public Int32 MVID_CR_CAM_SetImageOutPutMode_NET(
    MVID_IMGAE_OUTPUT_MODE  enImageOutPutMode
);
```

Parameters

enImageOutPutMode

Image output format. See **MVID_IMAGE_OUTPUT_MODE** for details.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.3.8 MVID_CR_CAM_GetOneFrameTimeout_NET

Get one picture in a blocking mode and process it by algorithm.

API Definition

```
public Int32 MVID_CR_CAM_GetOneFrameTimeout_NET(  
    IntPtr      pstFrameInfo,  
    UInt32      nMsec  
);
```

Parameters

pstFrameInfo

Image information

nMsec

Waiting timeout, unit: millisecond. An error code will be returned if no image is acquired within the timeout.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.3.9 MVID_CR_CAM_GetImageBuffer_NET

Get one original picture from the camera in a blocking mode.

API Definition

```
public Int32 MVID_CR_CAM_GetImageBuffer_NET(  
    ref MVID_IMAGE_INFO pstImageInfo,  
    UInt32      nMsec  
);
```

Parameters

pImageInfo

Image information. See ***MVID_IMAGE_INFO*** for details.

nMsec

Waiting timeout, unit: millisecond. An error code will be returned if no image is acquired when exceeding the timeout.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

Remarks

The API is invalid after calling ***MVID_CR_CAM_GetOneFrameTimeout_NET***.

3.3.10 MVID_CR_CAM_RegisterPreImageCallBack_NET

Register callback function for preprocessing image data.

API Definition

```
public Int32 MVID_CR_CAM_RegisterPreImageCallBack_NET(  
    cbPreOutputdelegate      cbPreOutput,  
    IntPtr                  pUser  
);
```

Parameters

cbPreOutput

Callback function of image data preprocessing

pUser

User pointer

Return Value

Return *0* for success, and return ***Error Code*** for failure.

Remarks

The format of input images should be mono8 after preprocessing.

3.3.11 MVID_CR_RegisterExceptionCallBack_NET

Register callback function for getting exception information of cameras and dongles.

API Definition

```
public Int32 MVID_CR_RegisterExceptionCallBack_NET(  
    cbExceptiondelegate      cbException,  
    IntPtr                  pUser  
);
```


Parameters

cbException

Exception information callback function

pUser

User pointer

Return Value

Return *0* for success, and return ***Error Code*** for failure.

Remarks

You can get exception information via this API without linking a camera.

3.3.12 MVID_CR_SaveImage_NET

Save the image in BMP or JPEG format with the encoding quality between 50 and 99.

API Definition

```
public Int32 MVID_CR_SaveImage_NET(  
    ref MVID_IMAGE_INFO    pstInputImage,  
    MVID_IMAGE_TYPE        enImageType,  
    ref MVID_IMAGE_INFO    pstOutputImage,  
    UInt32                  nJpgQuality  
);
```

Parameters

pstInputImage

Input image parameters. See ***MVID_IMAGE_INFO*** for details.

enImageType

The format to convert to, which is JPEG by default. See ***MVID_IMAGE_TYPE*** for details.

nJpgQuality

JPG encoding quality. It is valid only when **enImageType** is "JPEG".

pstOutputImage

Output image parameters. See ***MVID_IMAGE_INFO*** for details.

Return Value

Return *0* for success, and return ***Error Code*** for failure.

3.3.13 MVID_CR_GetImageData_NET

Read the image and convert it to mono8 format, which can be used in MVID_CR_Process_NET.

API Definition

```
public int32 MVID_CR_GetImageData_NET(  
    String          pFilePath  
    IntPtr          pstImageParam  
);
```

Parameters

pFilePath

Path of the input image file.

pstImageParam

Output image parameters.

Return Value

Return 0 for success, and return ***Error Code*** for failure.

Remarks

Supports images in RAW, BMP, JPEG, format. For RAW images, the input of width, height, and pixel format is required.

3.3.14 MVID_CR_RuleLoad_NET

Import filter rule files.

API Definition

```
public Int32 MVID_CR_RuleLoad_NET(  
    String          pFileName  
);
```

Parameters

pFileName

Path of the filter rule file. You can set it to NULL to cancel the filter.

Return Value

Return 0 for success, and return ***Error Code*** for failure.

Remarks

Filter rule files are generated by the tool RuleFilter.

3.3.15 MVID_CR_ScriptLoad_NET

Import script files.

API Definition

```
public Int32 MVID_CR_ScriptLoad_NET(  
    String      pFilePath,  
    String      pFuncName  
);
```

Parameters

pFilePath

Path of the script file

pFuncName

Filter function name

Return Value

Return 0 for success, and return **Error Code** for failure.

3.3.16 MVID_CR_StopGrabbing_NET

Stop image acquisition.

API Definition

```
public Int32 MVID_CR_StopGrabbing_NET(  
);
```

Return Value

Return 0 for success, and return **Error Code** for failure.

3.4 Algorithm Settings APIs

3.4.1 MVID_CR_Algorithm_SetIntValue_NET

Set algorithm parameters of type int.

API Definition

```
public Int32 MVID_CR_Algorithm_SetIntValue_NET(  
    String      strParamKeyName,  
    Int32      nValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

nValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

3.4.2 MVID_CR_Algorithm_GetIntValue_NET

Get algorithm parameters of type int.

API Definition

```
public Int32 MVID_CR_Algorithm_GetIntValue_NET(  
    String      strParamKeyName,  
    ref Int32   pnValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

pnValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

3.4.3 MVID_CR_Algorithm_SetFloatValu_NET

Set algorithm parameters of type float.

API Definition

```
public Int32 MVID_CR_Algorithm_SetFloatValue_NET(  
    String      strParamKeyName,  
    Single      fValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

fValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

3.4.4 MVID_CR_Algorithm_GetFloatValue_NET

Get algorithm parameters of type float.

API Definition

```
public Int32 MVID_CR_Algorithm_GetFloatValue_NET(  
    String      strParamKeyName,  
    ref Single   pfValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

pfValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

3.4.5 MVID_CR_Algorithm_SetStringValue_NET

Set algorithm parameters of type string.

API Definition

```
public Int32 MVID_CR_Algorithm_SetStringValue_NET(  
    String    strParamKeyName,  
    String    strValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

pValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

3.4.6 MVID_CR_Algorithm_GetStringValue_NET

Get algorithm parameters of type string.

API Definition

```
public Int32 MVID_CR_Algorithm_GetFloatValue_NET(  
    String    strParamKeyName,  
    ref byte  pstrValue  
);
```

Parameters

strParamKeyName

Parameter name, which should be obtained from the Machine Vision Software according to the camera type. For the parameter description, refer to [***Algorithm Parameters***](#).

pstrValue

Parameter value

Return Value

Return 0 for success, and return [***Error Code***](#) for failure.

Chapter 4 Data Structure and Enumeration

4.1 Data Structure

4.1.1 MVID_CAM_ENUMVALUE

Structure about values of type enum.

Member	Data Type	Description
nCurValue	UInt32	Current value
nSupportedNum	UInt32	The number of data records
nSupportValue	Array of UInt32	Supported enumeration types, each array indicates one type, the maximum size is 64 bits.
nReserved	Array of UInt32	Reserved.

4.1.2 MVID_CAM_FLOATVALUE

Structure about values of type float.

Member	Data Type	Description
fCurValue	Single	Current value
fMax	Single	The maximum value
fMin	Single	The minimum value
nReserved	Array of UInt32	Reserved.

4.1.3 MVID_CAM_INTVALUE_EX

Structure about values of type Int64.

Member	Data Type	Description
nCurValue	Int64	Current value
nMax	Int64	The maximum value

Member	Data Type	Description
nMin	Int64	The minimum value
nInc	Int64	Increment
nReserved	Array of UInt32	Reserved.

4.1.4 MVID_CAM_OUTPUT_INFO

Structure about output image information.

Member	Data Type	Description
stImage	<u>MVID_IMAGE_INFO</u>	Image information
stCodeList	<u>MVID_CODE_INFO_LIST</u>	Barcode information
pImageWaybill	IntPtr	Image matting buffer
nImageWaybillLen	UInt32	Image size
enWaybillImageType	<u>MVID_IMAGE_TYPE</u>	Image format
nReserved	Array of UInt32	Reserved.

4.1.5 MVID_CAM_STRINGVALUE

Structure about values of type string.

Member	Data Type	Description
chCurValue	Array of string	Current value
nMaxLength	Int64	The maximum size
nReserved	Array of UInt32	Reserved.

4.1.6 MVID_CAMERA_INFO

Device information structure.

Member	Data Type	Description
nCamType	UInt32	Camera type: USB, GigE
chManufacturerName	Array of String	Manufacturer name
chModelName	Array of String	Device model
chDeviceVersion	Array of String	Device version No.
chManufacturerSpecificInfo	Array of String	Manufacturer's Specific information
chSerialNumber	Array of String	Device serial No.
chUserDefinedName	Array of Byte	Custom name
nMacAddrHigh	UInt32	High MAC address
nMacAddrLow	UInt32	Low MAC address
nCommomReaserved	Array of UInt32	Reserved
nCurrentIp	UInt32	Current IP address
nNetExport	UInt32	IP Address of GigE camera
nGigEReserved	Array of UInt32	Reserved for GigE camera
CrtlInEndPoint	Byte	Control input endpoint
CrtlOutEndPoint	Byte	Control output endpoint
StreamEndPoint	Byte	Stream endpoint
EventEndPoint	Byte	Event endpoint
idVendor	UInt16	Vendor ID
idProduct	UInt16	Product ID
nDeviceNumber	UInt32	Device ID
nUsbReserved	Array of UInt32	Reserved for USB camera
bSelectDevice	Boolean	Whether the camera belongs to the target model: true, false.
nReserved	Array of UInt32	Reserved.

4.1.7 MVID_CAMERA_INFO_LIST

Device information list structure.

Member	Data Type	Description
nCamNum	UInt32	The number of online devices
pstCamInfo	Array of IntPtr	Device information, up to 256 devices can be supported.

4.1.8 MVID_CODE_INFO

Bar code information structure.

Member	Data Type	Description
strCode	Array of string	Character
nLen	Int32	Character size
enBarType	<u>MVID_CODE_TYPE</u>	Barcode type
stCornerPt	Array of <u>MVID_POINT_I</u>	Barcode location
nAngle	Int32	Barcode angle, range: [0, 3600°]
nFilterFlag	Int32	Filter identifier: 0- normal code, 1-filter code
nReserved	Array of UInt32	Reserved.

4.1.9 MVID_CODE_INFO_LIST

Structure about bar code information list.

Member	Data Type	Description
nCodeNum	Int32	The number of bar codes
stCodeInfo	Array of <u>MVID_CODE_INFO</u>	Bar code information
nReserved	Array of UInt32	Reserved.

4.1.10 MVID_EVENT_OUT_INFO

Event callback information

Member	Data Type	Description
EventName	string	Event name
nEventID	UInt16	Event ID
nStreamChannel	UInt16	Stream channel ID
nBlockIdHigh	UInt32	High-bit of frame ID
nBlockIdLow	UInt32	Low-bit of frame ID
nTimestampHigh	UInt32	High-bit of timestamp
nTimestampLow	UInt32	Low-bit of timestamp
pEventData	IntPtr	Event data
nEventDataSize	UInt32	Event data size
nReserved	Array of UInt32	Reserved.

4.1.11 MVID_IMAGE_INFO

Output image information structure.

Member	Data Type	Description
pImageBuf	IntPtr	Original image buffer
nImageLen	UInt32	Original image size
enImageType	<u>MVID_IMAGE_TYPE</u>	Image type
nWidth	UInt16	Image width
nHeight	UInt16	Image height
nFrameNum	UInt32	Frame No.
nDevTimeStampHigh	UInt32	Higher 32-bits of timestamp
nDevTimeStampLow	UInt32	Lower 32-bits of timestamp
nReserved	Array of UInt32	Reserved.

4.1.12 MVID_IMAGE_OUTPUT_MODE

Image output format structure.

Member	Macro Definition Value	Description
MVID_OUTPUT_NORMAL	0	Convert all non-mono8 images to BGR format
MVID_OUTPUT_RAW	1	Output in the original image format

4.1.13 MVID_POINT_I

Point coordinate information structure.

Member	Data Type	Description
nX	Int32	X-coordinate
nY	Int32	Y-coordinate

4.1.14 MVID_PROC_PARAM

Algorithm parameter information structure

Member	Data Type	Description
pImageBuf	IntPtr	Original image buffer
nImageLen	UInt32	Original image size
enImageType	<u>MVID_IMAGE_TYPE</u>	Image type
nWidth	UInt16	Image width
nHeight	UInt16	Image height
stCodeList	<u>MVID_CODE_INFO_LIST</u>	Barcode information
pImageWaybill	IntPtr	Matting buffer, which is allocated by SDK
nImageWaybillLen	UInt32	Image size
enWaybillImageType	<u>MVID_IMAGE_TYPE</u>	Image format
nReserved	Array of UInt32	Reserved.

4.2 Enumeration

4.2.1 MVID_CODE_TYPE

Enumeration about bar code types.

Member	Macro Definition Value	Description
MVID_CODE_NONE	0	No recognizable bar code
MVID_CODE_TDCR_DM	1	DM code
MVID_CODE_TDCR_QR	2	QR code
MVID_CODE_NONE	8	EAN8 code
MVID_CODE_BCR_UPCE	9	UPCE code
MVID_CODE_BCR_UPCA	12	UPCA code
MVID_CODE_BCR_EAN13	13	EAN13 code
MVID_CODE_BCR_ISBN13	14	ISBN13 code
MVID_CODE_BCR_CODABAR	20	Codabar code
MVID_CODE_BCR_ITF25	25	ITF25 code
MVID_CODE_BCR_CODE39	39	Code 39
MVID_CODE_BCR_CODE93	93	Code 93
MVID_CODE_BCR_CODE128	128	Code 128

4.2.2 MVID_IMAGE_TYPE

Picture saving format enumeration.

Member	Macro Definition Value
MVID_IMAGE_Undefined	0
MVID_IMAGE_MONO8	1
MVID_IMAGE_JPEG	2
MVID_IMAGE_BMP	3
MVID_IMAGE_RGB24	4
MVID_IMAGE_BGR24	5
MVID_IMAGE_MONO10	6
MVID_IMAGE_MONO10_Packed	7
MVID_IMAGE_MONO12	8
MVID_IMAGE_MONO12_Packed	9
MVID_IMAGE_MONO16	10
MVID_IMAGE_BayerGR8	11
MVID_IMAGE_BayerRG8	12
MVID_IMAGE_BayerGB8	13
MVID_IMAGE_BayerBG8	14
MVID_IMAGE_BayerGR10	15
MVID_IMAGE_BayerRG10	16
MVID_IMAGE_BayerGB10	17
MVID_IMAGE_BayerBG10	18
MVID_IMAGE_BayerGR12	19
MVID_IMAGE_BayerRG12	20
MVID_IMAGE_BayerGB12	21
MVID_IMAGE_BayerBG12	22
MVID_IMAGE_BayerGR10_Packed	23
MVID_IMAGE_BayerRG10_Packed	24
MVID_IMAGE_BayerGB10_Packed	25
MVID_IMAGE_BayerBG10_Packed	26
MVID_IMAGE_BayerGR12_Packed	27

Member	Macro Definition Value
MVID_IMAGE_BayerRG12_Packed	28
MVID_IMAGE_BayerGB12_Packed	29
MVID_IMAGE_BayerBG12_Packed	30
MVID_IMAGE_YUV422_Packed	31
MVID_IMAGE_YUV422_YUYV_Packed	32
MVID_IMAGE_RGB8_Packed	33
MVID_IMAGE_BGR8_Packed	34
MVID_IMAGE_RGBA8_Packed	35
MVID_IMAGE_BGRA8_Packed	36

4.2.3 MVID_IMAGE_OUTPUT_MODE

Image output format structure.

Member	Macro Definition Value	Description
MVID_OUTPUT_NORMAL	0	Convert all non-mono8 images to BGR format
MVID_OUTPUT_RAW	1	Output in the original image format

Appendix A. Appendixes

A.1 Algorithm Parameters

This page provides the description and value setting suggestions of algorithm parameters that are obtained from Machine Vision Software according to camera type.

One-Dimension Parameters

Parameter Name	Description	Value Range
KEY_BCR_ABILITY	Algorithm capability set, including Code39[1], Code128[2], CodeBar[4], EAN[8], ITF25[16], CODE93[32]	Range: [0, 63], default: 63
KEY_BCR_ROI_X	ROI offset in X direction	Range: [0, 65535], default: 0
KEY_BCR_ROI_Y	ROI offset in Y direction	Range: [0, 65535], default: 0
KEY_BCR_ROI_WIDTH	ROI width	Range: [100, 65535], default: 65535
KEY_BCR_ROI_HEIGHT	ROI height	Range: [40, 65535], default: 65535
KEY_BCR_MAX_WIDTH	The maximum width of images that can be recognized by the algorithm.	Range: [0, 65535], default: 3840
KEY_BCR_MAX_HEIGHT	The maximum height of images that can be recognized by the algorithm.	Range: [0, 65535], default: 2748
KEY_BCR_LOCBARNUM	The number of barcodes to read	Range: [1, 200], default: 4
KEY_BCR_LOCWINSIZE	Barcode height	Range: [4, 65], default: 4
KEY_BCR_WAITINGTIME	Algorithm library's maximum running time. The current image processing will be forced to stop when the specific time runs out.	Range: [0, 5000], default: 500
KEY_BCR_SEGQUIETW	Width of the barcode quiet zone	Range: [0, 200], default: 30

Parameter Name	Description	Value Range
KEY_BCR_DFKSIZELOWERLIMIT	Lower-limit of barcode width, below which the bar code will be filtered out.	Range: [0, 4000], default: 30
KEY_BCR_DFKSIZEUPPERLIMIT	Upper-limit of barcode width, above which the barcode will be filtered out.	Range: [0, 4000], default: 2400
KEY_BCR_SAVEIMAGELEVEL	Sensitivity of saving unanalyzed images	Range: [1, 3], default: 1
KEY_BCR_APPMODE	Algorithm library operating mode (expert mode or normal mode)	Range: [0, 2147483646], default: 1
KEY_BCR_DISTORTION	Whether to enable perspective distortion switch in expert mode	Range: [0, 2147483646], default: 0
KEY_BCR_WHITEGAP	Whether to enable printing quality switch in expert mode	Range: [0, 2147483646], default: 0
KEY_BCR_SPOT	Whether to enable specular reflective switch in expert mode	Range: [0, 2147483646], default: 0
KEY_BCR_SAMPLELEVEL	Image sampling scale	Range: [1, 8], default: 1
KEY_BCR_IAMGEMORPH	Image morphology preprocessing	Range: [0, 2], default: 0
KEY_BCR_DELERRFLAG	Whether to enable reducing the error identification	Range: [0, 1], default: 1

Two-Dimension Parameters

Parameter Name	Description	Value Range
KEY_TDCR_ABILITY	Algorithm capability set, including QR[1], DM[2]	Range: [0, 3], default: 3
KEY_TDCR_ROI_X	ROI offset in X direction	Range: [0, 65535], default: 0
KEY_TDCR_ROI_Y	ROI offset in Y direction	Range: [0, 65535], default: 0
KEY_TDCR_ROI_WIDTH	ROI width	Range: [128, 65535], default: 65535
KEY_TDCR_ROI_HEIGHT	ROI height	Range: [128, 65535], default: 65535

Parameter Name	Description	Value Range
KEY_TDCR_MAX_WIDTH	The maximum width of images that can be recognized by the algorithm.	Range: [0, 65535], default: 3840
KEY_TDCR_MAX_HEIGHT	The maximum height of images that can be recognized by the algorithm.	Range: [0, 65535], default: 2748
KEY_TDCR_LOCCODENUM	The number of ROIs outputted by detection module.	Range: [1, 1000], default: 5
KEY_TDCR_MINBARSIZE	The minimum width and height of barcodes when filtering blobs	Range: [20, 1000], default: 40
KEY_TDCR_MAXBARSIZE	The maximum width and height when filtering blobs	Range: [20, 1000], default: 300
KEY_TDCR_MIRRORMODE	Whether to enable mirror mode	Range: [0, 2], default: 0
KEY_TDCR_SAMPLELEVEL	Image downsampling ratio	Range: [1, 8], default: 1
KEY_TDCR_CODECOLOR	Identifier of the black bar code on white background	Range: [0, 2], default: 0
KEY_TDCR_DISCRETEFLAG	Code flag: "0"-continuous code, "1"-discrete code, "2"-self-adaptive	Range: [0, 2], default: 0
KEY_TDCR_DISTORTIONFLAG	QR distortion parameter	Range: [0, 1], default: 0
KEY_TDCR_WAITINGTIME	Exiting timeout	Range: [0, 5000], default: 1000
KEY_TDCR_DEBUGFLAG	Whether to display debug information	Range: [0, 1], default: 0
KEY_TDCR_ADVANCEPARAM	Advanced parameters	Range: [0, 2147483640], default: 0
KEY_TDCR_ADVANCEPARAM2	Advanced parameters 2	Range: [0, 2147483640], default: 0
KEY_TDCR_RECTANGLE	Code types: 0-square, 1-rectangle, 2-compatible mode	Range: [0, 2], default: 0
KEY_TDCR_APPMODE	Algorithm library operating mode (expert mode, normal mode, or speed mode)	Range: [0, 2], default: 0

Image Matting Parameters

Parameter Name	Description	Value Range
KEY_WAYBILL_ABILITY	Algorithm capability set	Range: [1, 7], default: 7
KEY_WAYBILL_MAX_WIDTH	The maximum width of images that can be recognized by the algorithm.	Range: [0, 65535], default: 3840
KEY_WAYBILL_MAX_HEIGHT	The maximum height of images that can be recognized by the algorithm.	Range: [0, 65535], default: 2748
KEY_WAYBILL_OUTPUTIMAGETYPE	Image output format	Range: [1, 3], default: 2 "1"-mono8, "2"-JPG, "3"-BMP
KEY_WAYBILL_JPGQUALITY	JPG encoding quality	Range: [1, 100], default: 80
KEY_WAYBILL_MINWIDTH	Minimum width of waybill	Range: [15, 2592], default: 100
KEY_WAYBILL_MINHEIGHT	Minimum height of waybill	Range: [10, 2048], default: 100
KEY_WAYBILL_MAXWIDTH	Maximum width of waybill	Range: [15, 3072], default: 3072
KEY_WAYBILL_MAXHEIGHT	Maximum height of waybill	Range: [10, 2048], default: 2048
KEY_WAYBILL_MORPHTIMES	Expansion times	Range: [0, 10], default: 0
KEY_WAYBILL_GRAYLOW	Minimum gray value of the bar code and character gray on the waybill	Range: [0, 255], default: 0
KEY_WAYBILL_GRAYMID	Median gray value of waybill which is used to distinguish barcode from background	Range: [0, 255], default: 70
KEY_WAYBILL_GRAYHIGH	Maximum gray value of waybill background	Range: [0, 255], default: 130
KEY_WAYBILL_BINARYADAPTIVE	Adaptive binarization	Range: [0, 1], default: 1
KEY_WAYBILL_BOUNDARYROW	Expand the edge in row direction when matting waybill	Range: [0, 2000], default: 0
KEY_WAYBILL_BOUNDARYCOL	Expand the edge in column direction when matting waybill	Range: [0, 2000], default: 0

Parameter Name	Description	Value Range
KEY_WAYBILL_MAXBILLBARHEIGTHRATIO	Maximum height ratio of waybill to barcode	Range: [1, 100], default: 20
KEY_WAYBILL_MAXBILLBARWIDTHRATIO	Maximum width ratio of waybill to barcode	Range: [1, 100], default: 5
KEY_WAYBILL_MINBILLBARHEIGTHRATIO	Minimum height ratio of waybill to barcode	Range: [1, 100], default: 5
KEY_WAYBILL_MINBILLBARWIDTHRATIO	Minimum width ratio of waybill to barcode	Range: [1, 100], default: 2
KEY_WAYBILL_ENHANCEMETHOD	Enhancement method	Range: [1, 4], default: 2
KEY_WAYBILL_ENHANCECLIPRATIOLOW	Increase the low threshold ratio of stretching	Range: [0, 100], default: 1
KEY_WAYBILL_ENHANCECLIPRATIOHIGH	Increase the high threshold ratio of stretching	Range: [0, 100], default: 99
KEY_WAYBILL_ENHANCECONTRASTFACTOR	Contrast ratio	Range: [1, 10000], default: 100
KEY_WAYBILL_ENHANCESHARPENFACTOR	Sharpness	Range: [0,10000], default: 0
KEY_WAYBILL_SHARPENKERNELSIZE	Size of sharpening filter core	Range: [3, 15], default: 3
KEY_WAYBILL_CODEBOUNDARYROW	Expand the edge in row direction when matting weight memo	Range: [0, 2000], default: 0
KEY_WAYBILL_CODEBOUNDARYCOL	Expand the edge in column direction when matting weight memo	Range: [0, 2000], default: 0

A.2 Error Code

You can search for the detailed error descriptions according to returned error codes or error names.

Error Code	Error Name	Description
0x00000000	MVID_CR_OK	Succeeded.
0x80000000	MVID_CR_E_HANDLE	Error or invalid handle
0x80000001	MVID_CR_E_SUPPORT	The function is not supported
0x80000002	MVID_CR_E_BUFOVER	Buffer is full
0x80000003	MVID_CR_E_CALLORDER	Incorrect calling order
0x80000004	MVID_CR_E_PARAMETER	Incorrect parameter
0x80000005	MVID_CR_E_RESOURCE	Applying resource failed.
0x80000006	MVID_CR_E_NODATA	No data
0x80000007	MVID_CR_E_PRECONDITION	Incorrect precondition, or the running environment changed.
0x80000008	MVID_CR_E_ENCRYPT	Credential error, possibly because the dongle was not installed or expired.
0x8000000A	MVID_CR_E_RULE	Filter rule error.
0x8000000B	MVID_CR_E_LOAD_LIBRARY	Dynamically importing the DLL file failed.
0x80000012	MVID_CR_E_JPGENC	JPG encoding error.
0x80000013	MVID_CR_E_IMAGE	Abnormal image. Incomplete image caused by packet loss or incorrect image format, width, or height.
0x80000014	MVID_CR_E_CONVERT	Format conversion error
0x800000FF	MVID_CR_E_UNKNOW	Unknown error
0x80000100	MVID_CR_E_GC_GENERIC	Generic error
0x80000101	MVID_CR_E_GC_ARGUMENT	Invalid parameter
0x80000102	MVID_CR_E_GC_RANGE	The value is out of range.
0x80000103	MVID_CR_E_GC_PROPERTY	Attribute error
0x80000104	MVID_CR_E_GC_RUNTIME	Running environment error
0x80000105	MVID_CR_E_GC_LOGICAL	Incorrect logic
0x80000106	MVID_CR_E_GC_ACCESS	Node accessing condition error
0x80000107	MVID_CR_E_GC_TIMEOUT	Timed out

Error Code	Error Name	Description
0x80000108	MVID_CR_E_GC_DYNAMICCAST	Conversion exception
0x800001FF	MVID_CR_E_GC_UNKNOW	GenICam unknown error
0x80000200	MVID_CR_E_NOT_IMPLEMENTED	Command is not supported by device.
0x80000201	MVID_CR_E_INVALID_ADDRESS	Target address does not exist
0x80000202	MVID_CR_E_WRITE_PROTECT	Target address is not writable.
0x80000203	MVID_CR_E_ACCESS_DENIED	No access permission
0x80000204	MVID_CR_E_BUSY	Device is busy, or network is disconnected.
0x80000205	MVID_CR_E_PACKET	Network packet error
0x80000206	MVID_CR_E_NETER	Network error
0x80000221	MVID_CR_E_IP_CONFLICT	IP address conflicted.
0x80000300	MVID_CR_E_USB_READ	USB read error
0x80000301	MVID_CR_E_USB_WRITE	USB write error
0x80000302	MVID_CR_E_USB_DEVICE	Device exception
0x80000303	MVID_CR_E_USB_GENICAM	GenICam error
0x80000304	MVID_CR_E_USB_BANDWIDTH	Insufficient bandwidth
0x80000305	MVID_CR_E_USB_DRIVER	Driver is mismatched, or is not installed.
0x800003FF	MVID_CR_E_USB_UNKNOW	USB unknown error
0x80002100	MVID_CR_E_CAMERA	Camera error
0x80002200	MVID_CR_E_BCR	1D barcode error
0x80002300	MVID_CR_E_TDCR	2D barcode error
0x80002400	MVID_CR_E_WAYBILL	Matting error
0x80002500	MVID_CR_E_SCRIPT	Script rule error

